

KryloCanvas (Web-WhiteBoard)

The Ultimate Interactive Whiteboard & Geometry Studio

Created by Abhimat Chatterjee

Welcome to the official, comprehensive instruction manual for KryloCanvas (Web-WhiteBoard)! Whether you are a student exploring complex geometry, a teacher explaining concepts, or an artist creating digital masterpieces, this guide will walk you through every single feature of the software.

🌟 1. Introduction: What is KryloCanvas (Web-WhiteBoard)?

KryloCanvas (Web-WhiteBoard) is a highly optimized, fully responsive web-based whiteboard application. It was built from the ground up to solve the frustrations of traditional digital drawing tools. Unlike basic whiteboards, KryloCanvas (Web-WhiteBoard) features a **Universal Styling Engine**, mathematically solid **Smart Geometry Tools**, and a robust **PDF/Image I/O Pipeline**.

Designed to work flawlessly on both desktop and mobile devices, the interface is split into two primary zones:

🔝 The Top Toolbar

Your command center. This horizontally scrollable bar houses your File Management (Import/Export), Undo/Redo commands, Theme toggles, and System controls (Refresh and Close).

👉 The Side/Bottom Palette

Your creative toolbox. This panel contains all your active drawing instruments: the Select Tool, Pen, Pencil, Highlighter, Erasers, Shapes Library, Rulers, and the Universal Color/Thickness controls.

🎨 2. The Universal Drawing Engine

KryloCanvas (Web-WhiteBoard) uses a synchronized state system for all drawing activities. This means your color and thickness choices apply universally across all relevant tools, saving you time and preventing styling mismatches.

🖌️ Pen & Pencil

Pen: Delivers bold, consistent strokes matching your exact chosen Universal Thickness.
Pencil: Simulates a real graphite pencil by automatically halving the Universal Thickness for finer, sharper sketching.

🖍️ Highlighter

Select any color from the color picker! The engine mathematically converts your solid hex color into a beautiful, semi-transparent layer. Strokes stack naturally over existing text and drawings without obscuring them.

🌈 Universal Sliders

Located at the bottom of your tool palette. Pick a color and slide the **Universal Thickness** bar. Whatever you do next—whether drawing a line or inserting a hexagon—will instantly adopt these exact settings.

🧽 Smart Erasers

Section Eraser: Click and drag a box over an area to delete entire objects instantly.
Pixel Eraser: Scribble directly over a specific stroke. If your eraser path intersects a line, only that specific line is deleted.

📐 3. Smart Geometry Tools & Snapping

KryloCanvas (Web-WhiteBoard) is heavily engineered for school-level geometry and technical drawing. Instead of just giving you static images of rulers, the software provides **Mathematically Solid Instruments**. When you draw near them, your pen will snap perfectly to their edges!

📏 The Solid Ruler

How to use: Click the *Rulers* icon and select the Ruler. It spawns on your canvas with dual markings (CM/MM on top, Inches on bottom).
Extensible: Drag the blue corner handles. The ruler dynamically generates new markings up to 100cm without stretching the text!
Edge Snapping: Select the Pen tool and draw along the ruler's edge. Your line will "magnetically" lock to the side, creating a perfectly straight line without bleeding through the solid plastic.

📐 Protractor & Set Square

Protractor: A semi-transparent 180° protractor with dual inner and outer degree markings. Trace the curved edge to snap perfect arcs!
Set Square: Perfect for drawing strict perpendicular and parallel lines. Like all instruments, it acts as a physical boundary for your pen.

🧭 Masterclass: Using the Interactive Compass

The KryloCanvas (Web-WhiteBoard) Compass is a true physical simulation, allowing you to accurately measure lengths and draw precise arcs.

- ✨ **Spawning:** Open the Rulers menu and select the Compass. It appears with two main control nodes: a 🔴 **Red Pivot Pin** and a 🔵 **Blue Pencil Node**.
- ✨ **Placing & Measuring:** Click and drag the 🔴 Red Pin to move the entire compass across the board. The distance between the pin and pencil remains locked, allowing you to measure a line against the ruler and carry that measurement across the page!
- ✨ **Drawing:** Click and drag the 🔵 Blue Pencil Node. It will strictly pivot around the Red Pin, leaving a perfect arc on the canvas in your current Universal Color and Thickness.
- ✨ **Removing:** Click the compass body to reveal the custom red ✖️ Delete button in the top right corner.

> 💡 **Pro Tip for Instruments: Need to move a ruler out of the way after drawing? Simply click the arrow "Select" tool. This instantly unlocks all instruments, allowing you to drag them, rotate them using the blue circular arrows, or delete them!**

💠 The 25-Shape Library

Click the **Shapes** icon (purple shapes) to open a beautiful visual grid of 25 geometric figures.

Category	Included Shapes
Triangles	Equilateral, Isosceles, Scalene, Right-Angled.
Quadrilaterals	Square, Rectangle, Parallelogram, Rhombus, Trapezium, Kite.
Polygons & Curves	Pentagon, Hexagon, Heptagon, Octagon, Circle, Semicircle, Oval.
Symbols	Star, Cross, Arrows (Right & Up), Heart, Moon, Cloud, Wedge.

How to Color Shapes: First, pick your color and thickness in the palette. *Then* click the shape you want. It will instantly spawn on the canvas perfectly matching your selections!

📥 4. File Management: Import & Export

KryloCanvas (Web-WhiteBoard) is deeply integrated with professional PDF and Image rendering pipelines. The Top Toolbar provides four distinct buttons for bringing your work in and getting it out.

🖼️ Import Image

Action: Loads a JPG or PNG onto your current active page.
Function: The image drops in as a fully interactive object. Switch to the *Select* tool to freely move it, scale it from the corners, or rotate it using the top rotation handle to fix the orientation!

📑 Import PDF

Action: Loads a multi-page PDF document into the workspace.
Function: KryloCanvas (Web-WhiteBoard) acts as a document editor! It analyzes the PDF and automatically generates a *brand new whiteboard page* for every single page of your PDF in sequential order. You can instantly begin drawing over the imported document.

📤 Export PDF

Action: Saves your entire workspace to your local device.

Function: The engine loops through every active page from top to bottom, hides the UI elements (like the semi-transparent rulers), and stitches the pages together into a pristine PDF document.

⏰ **Automated Naming Convention: You never have to worry about file organization! Every PDF you export is automatically named according to strict formatting rules with the current date. For example:** `KryloCanvas (Web-WhiteBoard)_13.10.2026.pdf`.

🌓 5. Canvas Operations & Page Management

The Multi-Page Workspace

KryloCanvas (Web-WhiteBoard) isn't limited to a single screen. Scroll down to the bottom of your workspace and click the large, dashed ➕ **ADD PAGE** button. A fresh canvas will be appended. Need to delete a page? Every page (except the last remaining one) has a prominent red ✖ button in the top right corner.

Millimeter Graphs

Beside the Theme options in the Top Toolbar is a green checkered button 🏁. Clicking this instantly spawns a vertically-aligned page pre-loaded with a mathematically accurate millimeter graph grid—perfect for plotting data or geometric tracing!

Themes & Color Inversion

In the Top Toolbar, you will find three circular theme buttons: **White**, **Black**, and **Grey**.

- ✨ When you switch from White to Black, KryloCanvas (Web-WhiteBoard) scans your entire document. Any black ink strokes will automatically invert to white so they remain perfectly visible against the dark background!
- ✨ Switching back to White or Grey will revert those strokes to black.
- ✨ The outer UI (toolbars) always remains in an optimized dark-mode to reduce eye strain, while only the paper changes color.

🔎 6. Navigation, Protection, & Pro Features

🔍 Native Browser Zoom

We removed clunky custom magnifying glasses! KryloCanvas (Web-WhiteBoard) fully supports your device's native zooming. On a desktop, hold **CTRL** and scroll your mouse wheel. On mobile, use a standard two-finger pinch. Because everything is vector-based, your drawings and rulers will remain crystal clear at any zoom level.

↩️ Responsive Undo/Redo

Located in the top toolbar, the Undo and Redo buttons track everything on a *per-page* basis. Whether you drew a line, changed a shape's color, or erased a section, the robust history stack ensures you never lose a good idea (or suffer from a bad one).

🛡️ Data Protection Modals

Accidents happen, so KryloCanvas (Web-WhiteBoard) protects you from them.

- Clicking **Clear All** asks for confirmation before wiping the canvas.
- Clicking the top-bar **Refresh** 🔄 button issues a strict warning that unsaved data will be lost.
- Clicking **Close** ✖️ asks: *"Confirm you have saved the document. Close without saving?"*

📱 Mobile 100dvh Layout

If you open KryloCanvas (Web-WhiteBoard) on your phone, you will notice the toolbars never get cut off by your browser's address bar. The app strictly enforces a 100dvh layout, automatically moving the tool palette to the bottom of the screen while keeping everything horizontally scrollable.

🎯 7. Quick Troubleshooting Checklist

Problem	Solution
I can't move the ruler!	You are likely in "Drawing Mode" (Pen/Pencil). Click the arrow **Select** tool in the side toolbar to unlock the ruler for moving or rotating.

Problem	Solution
My lines are jagged near the ruler.	Make sure you are drawing *close* to the ruler's edge. The magnetic snap engages when you get within a few pixels of the solid boundary.
I can't see all the tools on mobile.	The top and bottom toolbars are horizontally scrollable! Just swipe left or right on the buttons to reveal the hidden options.
Shapes are the wrong color.	The Universal Engine locks the color in the moment you click the shape. Change your color in the palette *first*, then click the shape.

Thank you for using KryloCanvas (Web-WhiteBoard)!

This software was meticulously designed to bring professional, robust, and mathematically accurate digital drawing tools to the web. Your workflow is now faster, smarter, and infinitely more creative.

🚀 Keep Creating!



KryloCanvas (Web-WhiteBoard)

The Final Frontier of Digital Whiteboarding

Software Architecture & Design by

Abhimat Chatterjee

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